

|                                          | $\mathcal{K}_{0+}^{\#1}$                                  | $\mathcal{K}_{0\beta X}^{\#0}$                            |                          |                          |                                          |                          |
|------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------|--------------------------|--------------------------|------------------------------------------|--------------------------|
| $\mathcal{K}_{0+}^{\#1} +$               | $\frac{\gamma_1 k^2}{a}$                                  | 0                                                         |                          |                          |                                          |                          |
| $\mathcal{K}_{0+}^{\#1} + \alpha\beta X$ | 0                                                         | $\gamma_2 k^2 + 3 \overset{(2)}{K_1}$                     | $\mathcal{K}_{1-}^{\#1}$ | $\mathcal{K}_{1-}^{\#2}$ | $\mathcal{K}_{1-}^{\#1}$                 | $\mathcal{K}_{1-}^{\#2}$ |
| $\mathcal{K}_{1-}^{\#1} + \alpha\beta$   | $\frac{1}{2} (\gamma_3 k^2 + 4 \overset{(2)}{K_1})$       | $\frac{-\gamma_4 k^2 + 4 \overset{(2)}{K_1}}{2 \sqrt{2}}$ | 0                        | 0                        | $\mathcal{K}_{2-}^{\#1}$                 | $\mathcal{K}_{2-}^{\#1}$ |
| $\mathcal{K}_{1-}^{\#2} + \alpha\beta$   | $\frac{-\gamma_4 k^2 + 4 \overset{(2)}{K_1}}{2 \sqrt{2}}$ | $\frac{1}{4} (-\gamma_5 k^2 + 4 \overset{(2)}{K_1})$      | 0                        | 0                        | $\mathcal{K}_{2-}^{\#1}$                 | $\mathcal{K}_{2-}^{\#1}$ |
| $\mathcal{K}_{1-}^{\#1} + \alpha$        | 0                                                         | 0                                                         | 0                        | 0                        | $\mathcal{K}_{2+}^{\#1}$                 | $\mathcal{K}_{2+}^{\#1}$ |
| $\mathcal{K}_{1-}^{\#2} + \alpha$        | 0                                                         | 0                                                         | 0                        | 0                        | $\mathcal{K}_{2+}^{\#1}$                 | $\mathcal{K}_{2+}^{\#1}$ |
|                                          |                                                           |                                                           |                          |                          | $\mathcal{K}_{2+}^{\#1} + \alpha\beta$   | $\mathcal{K}_{2+}^{\#1}$ |
|                                          |                                                           |                                                           |                          |                          | $\frac{\gamma_6 k^2}{2}$                 | 0                        |
|                                          |                                                           |                                                           |                          |                          | $\mathcal{K}_{2+}^{\#1} + \alpha\beta X$ | $\mathcal{K}_{2+}^{\#1}$ |
|                                          |                                                           |                                                           |                          |                          | 0                                        | $\frac{\gamma_7 k^2}{2}$ |

$$\begin{array}{c}
\begin{array}{cc}
\mathcal{J}_{0^+}^{\#1} \uparrow & \mathcal{J}_{0^+}^{\#0} \alpha\beta\chi \\
\begin{array}{|c|c|}
\hline
1 & 0 \\
\hline
\text{Det}(0^+) & \\
\hline
\end{array} & \begin{array}{|c|}
\hline
0 \\
\hline
\end{array} \\
\mathcal{J}_{0^+}^{\#1} \uparrow \alpha\beta\chi & \begin{array}{|c|}
\hline
1 \\
\hline
\text{Det}(0^+) \\
\hline
\end{array}
\end{array} \\
\begin{array}{cccc}
\mathcal{J}_{1^+}^{\#1} \uparrow \alpha\beta & \begin{array}{|c|}
\hline
-\gamma_3 k^2 + 4 \frac{(2)}{k_+} \\
\hline
4 \text{Det}(1^+) \\
\hline
\end{array} & \begin{array}{|c|}
\hline
\gamma_4 k^2 - 4 \frac{(2)}{k_+} \\
\hline
2 \sqrt{2} \text{Det}(1^+) \\
\hline
\end{array} & \begin{array}{|c|}
\hline
0 \\
\hline
\end{array} \\
\mathcal{J}_{1^+}^{\#2} \uparrow \alpha\beta & \begin{array}{|c|}
\hline
\gamma_4 k^2 - 4 \frac{(2)}{k_+} \\
\hline
2 \sqrt{2} \text{Det}(1^+) \\
\hline
\end{array} & \begin{array}{|c|}
\hline
\gamma_3 k^2 + 4 \frac{(2)}{k_+} \\
\hline
2 \text{Det}(1^+) \\
\hline
\end{array} & \begin{array}{|c|}
\hline
0 \\
\hline
\end{array} \\
\mathcal{J}_{1^+}^{\#1} \uparrow \alpha & \begin{array}{|c|}
\hline
0 \\
\hline
\end{array} & \begin{array}{|c|}
\hline
0 \\
\hline
\end{array} & \begin{array}{|c|}
\hline
0 \\
\hline
\end{array} \\
\mathcal{J}_{1^+}^{\#2} \uparrow \alpha & \begin{array}{|c|}
\hline
0 \\
\hline
\end{array} & \begin{array}{|c|}
\hline
0 \\
\hline
\end{array} & \begin{array}{|c|}
\hline
0 \\
\hline
\end{array}
\end{array}
\end{array}
\begin{array}{cc}
\mathcal{J}_{2^+}^{\#1} \alpha\beta & \mathcal{J}_{2^+}^{\#1} \alpha\beta\chi \\
\begin{array}{|c|}
\hline
1 \\
\hline
\text{Det}(2^+) \\
\hline
\end{array} & \begin{array}{|c|}
\hline
0 \\
\hline
\end{array} \\
\mathcal{J}_{2^+}^{\#1} \uparrow \alpha\beta\chi & \begin{array}{|c|}
\hline
0 \\
\hline
\end{array} \\
\mathcal{J}_{2^+}^{\#1} \uparrow \alpha\beta\chi & \begin{array}{|c|}
\hline
1 \\
\hline
\text{Det}(2^+) \\
\hline
\end{array}
\end{array}
\end{array}$$

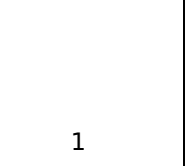
| Abbreviations used in matrices                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| $Y_1 = 6 \overset{(4)}{\kappa}_1 + 2 \overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16} + 2 \overset{(4)}{\kappa}_3 \overset{(4)}{\kappa}_6$ & $Y_2 = \overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_{16} \overset{(4)}{\kappa}_3$ & $Y_3 = 2 \overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_{16} - \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6$ & $Y_4 = 2 \overset{(4)}{\kappa}_{16} - \overset{(4)}{\kappa}_4 \overset{(4)}{\kappa}_6$ & $Y_5 = 2 \overset{(4)}{\kappa}_{15} + 3 \overset{(4)}{\kappa}_{16} - 2 \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4 \overset{(4)}{\kappa}_6$ & $Y_6 = 2 \overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16} - \overset{(4)}{\kappa}_3 - \overset{(4)}{\kappa}_6 \overset{(4)}{\kappa}_6$ & $Y_7 = 2 \overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16} \overset{(4)}{\kappa}_6$ & $\text{Det}(0^+) = -\frac{1}{4} k^2 (6 \overset{(4)}{\kappa}_1 + 2 \overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16} + 2 \overset{(4)}{\kappa}_3 \overset{(4)}{\kappa}_6)$ & $\text{Det}(1^+) = -\frac{1}{8} (-k^4 (4 \overset{(4)}{\kappa}_{15}^2 + \overset{(4)}{\kappa}_{16}^2 + 2 \overset{(4)}{\kappa}_3^2 + 2 \overset{(4)}{\kappa}_3 \overset{(4)}{\kappa}_4 + \overset{(4)}{\kappa}_4^2 - \overset{(4)}{\kappa}_{16} (\overset{(4)}{\kappa}_3 + 2 \overset{(4)}{\kappa}_4 - 3 \overset{(4)}{\kappa}_6) - 2 (\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4) \overset{(4)}{\kappa}_6 + 2 \overset{(4)}{\kappa}_{15} (2 \overset{(4)}{\kappa}_{16} - 3 \overset{(4)}{\kappa}_3 - 2 \overset{(4)}{\kappa}_4 + \overset{(4)}{\kappa}_6)) + 4 k^2 (\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6) \overset{(2)}{\kappa}_1)$ & $\text{Det}(2^+) = \frac{1}{2} k^2 (2 \overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16} - \overset{(4)}{\kappa}_3 - \overset{(4)}{\kappa}_6)$ |  |

$$\begin{aligned} & \text{Lagrangian} \\ & \binom{2}{1} \mathcal{K}_{a\beta X} \mathcal{K}^{a\beta X} - 2 \binom{2}{1} \mathcal{K}^{a\beta X} \mathcal{K}_{\beta aX} + \binom{4}{1} \partial_\beta \mathcal{K}^\delta_{\chi\delta} \partial^\chi \mathcal{K}^\alpha_a{}^\beta + \binom{4}{1} \partial_\chi \mathcal{K}^\delta_{\beta\delta} \partial^\chi \mathcal{K}^\alpha_a{}^\beta + \\ & \frac{1}{2} \binom{4}{1} \partial_\chi \mathcal{K}^\delta_{\beta\delta} \partial^\chi \mathcal{K}^\alpha_a{}^\beta + \binom{4}{1} \partial_\beta \mathcal{K}^{a\beta X} \partial_\delta \mathcal{K}^\delta_{\alpha X} + \binom{4}{1} \partial_\alpha \mathcal{K}^{a\beta X} \partial_\delta \mathcal{K}^\delta_{\beta X} + \binom{4}{1} \partial_\beta \mathcal{K}^{a\beta X} \partial_\delta \mathcal{K}^\delta_{\chi\alpha} - \\ & 2 \binom{4}{1} \partial^\chi \mathcal{K}^\alpha_a{}^\beta \partial_\delta \mathcal{K}^\delta_{\chi\beta} - \binom{4}{1} \partial^\chi \mathcal{K}^\alpha_a{}^\beta \partial_\delta \mathcal{K}^\delta_{\chi\beta} - \frac{3}{2} \binom{4}{1} \partial_\alpha \mathcal{K}^{a\beta X} \partial_\delta \mathcal{K}^\delta_{\beta X} - \frac{3}{4} \binom{4}{1} \partial_\alpha \mathcal{K}^{a\beta X} \partial_\delta \mathcal{K}^\delta_{\beta X} + \\ & \frac{1}{2} \binom{4}{1} \partial_\alpha \mathcal{K}^{a\beta X} \partial_\delta \mathcal{K}^\delta_{\beta X} + \frac{1}{2} \binom{4}{1} \partial_\alpha \mathcal{K}^{a\beta X} \partial_\delta \mathcal{K}^\delta_{\beta X} + \binom{4}{1} \partial_\delta \mathcal{K}^{a\beta X} \partial^\delta \mathcal{K}^{a\beta X} + \binom{4}{1} \partial_\delta \mathcal{K}^{a\beta X} \partial^\delta \mathcal{K}^{a\beta X} \end{aligned}$$

|                                 |                 |                                                                                                                                                                                                                |
|---------------------------------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Added source term(s):           |                 | $\mathcal{K}^{a\beta\chi} \mathcal{T}_{a\beta\chi}$                                                                                                                                                            |
| Source constraint(s)            | # constraint(s) | Covariant form                                                                                                                                                                                                 |
| $\mathcal{T}_1^{\#2\alpha} = 0$ | 3               | $\partial_\chi \partial_\beta \mathcal{T}^{\beta\alpha\chi} = 0$                                                                                                                                               |
| $\mathcal{T}_1^{\#1\alpha} = 0$ | 3               | $\partial_\chi \partial^\alpha \mathcal{T}_\beta^{\phantom{\alpha}\chi} + \partial_\chi \partial^\chi \mathcal{T}_\beta^{\phantom{\alpha}\alpha} = \partial_\chi \partial_\beta \mathcal{T}^{\beta\alpha\chi}$ |
| Total # constraint(s):          | 6               |                                                                                                                                                                                                                |

| Resolved pole(s) | # polarization(s) | Square mass | Residue                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------|-------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  | 1                 | 0           | $\frac{-4 \binom{(4)}{\kappa_{15}} 2 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_3} \binom{(4)}{\kappa_6}}{(2 \binom{(4)}{\kappa_{15} + \kappa_{16}} (2 \binom{(4)}{\kappa_{15}} \binom{(4)}{\kappa_{16}} \kappa_3 \kappa_6))}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                  | 1                 | 0           | $-\frac{-4 \binom{(4)}{\kappa_{15}} 2 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_3} \binom{(4)}{\kappa_6}}{(2 \binom{(4)}{\kappa_{15} + \kappa_{16}} (2 \binom{(4)}{\kappa_{15}} \binom{(4)}{\kappa_{16}} \kappa_3 \kappa_6))}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                  | 2                 | 0           | $\frac{28 \binom{(4)}{\kappa_{15}}^2 + 7 \binom{(4)}{\kappa_{16}}^2 - 8 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_3 + \kappa_6} + 4 \binom{(4)}{\kappa_3} \binom{(4)}{\kappa_6}^2 + 4 \binom{(4)}{\kappa_{15}} (7 \binom{(4)}{\kappa_{16}} - 4 \binom{(4)}{\kappa_3 + \kappa_6}))}{(2 \binom{(4)}{\kappa_{15} + \kappa_{16}} (2 \binom{(4)}{\kappa_{15}} \binom{(4)}{\kappa_{16}} \kappa_3 \kappa_6) (\kappa_3 + \kappa_6))}$                                                                                                                                                                                                                                                                        |
|                  | 1                 | 0           | $\frac{2 \binom{(4)}{\kappa_{15}} (\binom{(4)}{\kappa_3 + \kappa_6})^2 \binom{(4)}{\kappa_1} + \kappa_{16} \binom{(4)}{\kappa_3 + \kappa_6}^2 \binom{(4)}{\kappa_1} - \sqrt{\binom{(4)}{\kappa_3} \binom{(4)}{\kappa_6}}^4 (16 \binom{(4)}{\kappa_{15}}^2 + 4 \binom{(4)}{\kappa_{16}}^2 + 4 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_3} \binom{(4)}{\kappa_6} - 2 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_3 + \kappa_6} + (\binom{(4)}{\kappa_3} \binom{(4)}{\kappa_6})^2) \binom{(2)}{\kappa_1}^2}{(2 \binom{(4)}{\kappa_{15} + \kappa_{16}} (2 \binom{(4)}{\kappa_{15}} \binom{(4)}{\kappa_{16}} \kappa_3 \kappa_6) (\binom{(4)}{\kappa_3} \binom{(4)}{\kappa_6})^2 \binom{(2)}{\kappa_1})}$ |
|                  | 1                 | 0           | $\frac{2 \binom{(4)}{\kappa_{15}} (\binom{(4)}{\kappa_3 + \kappa_6})^2 \binom{(4)}{\kappa_1} + \kappa_{16} \binom{(4)}{\kappa_3 + \kappa_6}^2 \binom{(4)}{\kappa_1} + \sqrt{\binom{(4)}{\kappa_3} \binom{(4)}{\kappa_6}}^4 (16 \binom{(4)}{\kappa_{15}}^2 + 4 \binom{(4)}{\kappa_{16}}^2 + 4 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_3} \binom{(4)}{\kappa_6} - 2 \binom{(4)}{\kappa_{16}} \binom{(4)}{\kappa_3 + \kappa_6} + (\binom{(4)}{\kappa_3} \binom{(4)}{\kappa_6})^2) \binom{(2)}{\kappa_1}^2}{(2 \binom{(4)}{\kappa_{15} + \kappa_{16}} (2 \binom{(4)}{\kappa_{15}} \binom{(4)}{\kappa_{16}} \kappa_3 \kappa_6) (\binom{(4)}{\kappa_3} \binom{(4)}{\kappa_6})^2 \binom{(2)}{\kappa_1})}$ |

[illegible]



$$\begin{aligned}
& 27 \kappa_{16}^6 p^4 - 36 \kappa_{16}^5 (4 \kappa_3 + 3 \kappa_4 - 2 \kappa_6) p^4 + 576 \kappa_{15}^5 (9 \kappa_{16} - 8 \kappa_3 - 4 \kappa_4 + 4 \kappa_6) p^4 + \\
& 6 \kappa_{16}^4 (52 \kappa_3^2 + 27 \kappa_4^2 + 8 \kappa_3 (9 \kappa_4 - 5 \kappa_6) - 36 \kappa_4 \kappa_6 + 16 \kappa_6^2) p^4 + 48 \kappa_{15}^4 (135 \kappa_{16}^2 + \\
& 104 \kappa_3^2 + 54 \kappa_4^2 + 16 \kappa_3 (9 \kappa_4 - 5 \kappa_6) - 60 \kappa_{16} (4 \kappa_3 + 3 \kappa_4 - 2 \kappa_6) - 72 \kappa_4 \kappa_6 + 32 \kappa_6^2) p^4 - \\
& 4 \kappa_{16}^3 (80 \kappa_3^3 + 27 \kappa_4^3 + 12 \kappa_3^2 (13 \kappa_4 - 6 \kappa_6) - 54 \kappa_4^2 \kappa_6 + 48 \kappa_4 \kappa_6^2 - 16 \kappa_6^3) + \\
& 12 \kappa_3 (9 \kappa_4^2 - 10 \kappa_4 \kappa_6 + 4 \kappa_6^2) p^4 + 32 \kappa_{15}^3 (135 \kappa_{16}^3 - 80 \kappa_3^3 - 27 \kappa_4^3 - \\
& 90 \kappa_{16}^2 (4 \kappa_3 + 3 \kappa_4 - 2 \kappa_6) + 54 \kappa_4^2 \kappa_6 - 48 \kappa_4 \kappa_6^2 + 16 \kappa_6^3 + \kappa_3^2 (-156 \kappa_4 + 72 \kappa_6) - 12 \kappa_3^2 \kappa_4 \\
& (9 \kappa_4^2 - 10 \kappa_4 \kappa_6 + 4 \kappa_6^2) + 6 \kappa_{16}^2 (52 \kappa_3^2 + 27 \kappa_4^2 + 8 \kappa_3 (9 \kappa_4 - 5 \kappa_6) - 36 \kappa_4 \kappa_6 + 16 \kappa_6^2)) p^4 + \\
& \kappa_{16}^2 (128 \kappa_3^4 + 27 \kappa_4^4 + 64 \kappa_3^3 (5 \kappa_4 - 2 \kappa_6) p^4 - 72 \kappa_4^3 \kappa_6 p^4 - 64 \kappa_4 \kappa_6^3 p^4 + \\
& 16 \kappa_4^4 p^4 + 12 \kappa_6^2 (9 \kappa_1^2 + 8 \kappa_4^2 - 8 \kappa_3 (-27 \kappa_6 \kappa_1 - 18 \kappa_4^3 p^4 + 30 \kappa_4^2 \kappa_6 p^4 - \\
& 24 \kappa_4 \kappa_6^2 p^4 + 8 \kappa_6^3 p^4) + 12 \kappa_3^2 (9 \kappa_1^2 + 2 (13 \kappa_4^2 - 12 \kappa_4 \kappa_6 + 4 \kappa_6^2) p^4)) + \\
& 4 \kappa_{15}^2 (128 \kappa_3^4 p^4 + 27 \kappa_{16}^2 (15 \kappa_4^2 - 10 \kappa_{16} \kappa_4 + \kappa_4^2) p^4 + 72 \kappa_{16}^3 (-18 \kappa_{16} \kappa_4 + \\
& 9 \kappa_{16} \kappa_4^2 - \kappa_4^3) \kappa_6 p^4 + 64 (3 \kappa_{16}^2 - \kappa_4) \kappa_6^3 p^4 + 16 \kappa_4^4 p^4 - 64 \kappa_4^3 (15 \kappa_{16} - 5 \kappa_4 + 2 \kappa_6) p^4 + \\
& 12 \kappa_6^2 (9 \kappa_1^2 + 8 (6 \kappa_{16}^2 - 6 \kappa_{16} \kappa_4 + \kappa_4^2) p^4) - 8 \kappa_3 (-27 \kappa_6 \kappa_1 + 18 (10 \kappa_{16}^3 - 18 \kappa_{16}^2 \kappa_4 + \\
& 9 \kappa_{16} \kappa_4^2 - \kappa_4^3) p^4 + 30 (6 \kappa_{16}^2 - 6 \kappa_{16} \kappa_4 + \kappa_4^2) \kappa_6 p^4 + 24 (3 \kappa_{16}^2 - \kappa_4) \kappa_6^2 p^4 + 8 \kappa_6^3 p^4) + \\
& 12 \kappa_3^2 (9 \kappa_1^2 + 2 (78 \kappa_{16}^2 - 78 \kappa_{16} \kappa_4 + 13 \kappa_4^2 + 36 \kappa_{16} \kappa_6 - 12 \kappa_4 \kappa_6 + 4 \kappa_6^2) p^4)) + \\
& 4 \kappa_{15}^4 (36 (\kappa_3 + \kappa_6)^3 \kappa_{16}^2 + 81 \kappa_{16}^5 p^4 - 90 \kappa_{16}^4 (4 \kappa_3 + 3 \kappa_4 - 2 \kappa_6) p^4 + \\
& 12 \kappa_{16}^3 (52 \kappa_3^2 + 27 \kappa_4^2 + 8 \kappa_3 (9 \kappa_4 - 5 \kappa_6) - 36 \kappa_4 \kappa_6 + 16 \kappa_6^2) p^4 - 6 \kappa_{16}^2 (80 \kappa_3^3 + 27 \kappa_4^3 + \\
& 12 \kappa_3^2 (13 \kappa_4 - 6 \kappa_6) - 54 \kappa_4^2 \kappa_6 + 48 \kappa_4 \kappa_6^2 - 16 \kappa_6^3 + 12 \kappa_3 (9 \kappa_4^2 - 10 \kappa_4 \kappa_6 + 4 \kappa_6^2)) p^4 + \\
& \kappa_{16}^4 (128 \kappa_3^4 p^4 + 27 \kappa_4^4 p^4 + 64 \kappa_3^3 (5 \kappa_4 - 2 \kappa_6) p^4 - 72 \kappa_4^3 \kappa_6 p^4 - 64 \kappa_4 \kappa_6^3 p^4 + 16 \kappa_4^4 p^4 + \\
& 12 \kappa_6^2 (9 \kappa_1^2 + 8 \kappa_4^2 - 8 \kappa_3 (-27 \kappa_6 \kappa_1 - 18 \kappa_4^3 p^4 + 30 \kappa_4^2 \kappa_6 p^4 - 24 \kappa_4 \kappa_6^2 p^4 + 8 \kappa_6^3 p^4) + \\
& 12 \kappa_3^2 (9 \kappa_1^2 + 2 (13 \kappa_4^2 - 12 \kappa_4 \kappa_6 + 4 \kappa_6^2) p^4))))) / \\
& ((2 \kappa_{15}^4 + \kappa_{16}^4) (2 \kappa_{15}^4 + \kappa_{16}^4 - \kappa_3^2 - \kappa_6^2) (\kappa_3 + \kappa_6)^2 \kappa_{11}^2)
\end{aligned}$$

[illegible]



$$\begin{aligned}
& 27 \kappa_{16}^6 p^4 - 36 \kappa_{16}^5 (4 \kappa_4^{(4)} + 3 \kappa_4^{(4)} - 2 \kappa_6^{(4)}) p^4 + 576 \kappa_{15}^{(4)5} (9 \kappa_{16}^{(4)} - 8 \kappa_3^{(4)} - 6 \kappa_4^{(4)} + 4 \kappa_6^{(4)}) p^4 + \\
& 6 \kappa_{16}^{(4)4} (52 \kappa_3^{(4)2} + 27 \kappa_4^{(4)2} + 8 \kappa_3^{(4)} (9 \kappa_4^{(4)} - 5 \kappa_6^{(4)}) - 36 \kappa_4^{(4)} \kappa_6^{(4)} + 16 \kappa_6^{(4)2}) p^4 + 48 \kappa_{15}^{(4)4} (135 \kappa_{16}^{(4)2} + \\
& 104 \kappa_3^{(4)2} + 54 \kappa_4^{(4)2} + 16 \kappa_3^{(4)} (9 \kappa_4^{(4)} - 5 \kappa_6^{(4)}) - 60 \kappa_{16}^{(4)} (4 \kappa_3^{(4)} + 3 \kappa_4^{(4)} - 2 \kappa_6^{(4)}) - 72 \kappa_4^{(4)} \kappa_6^{(4)} + 32 \kappa_6^{(4)2}) p^4 - \\
& 4 \kappa_{16}^{(4)3} (80 \kappa_3^{(4)3} + 27 \kappa_4^{(4)3} + 12 \kappa_3^{(4)2} (13 \kappa_4^{(4)} - 6 \kappa_6^{(4)}) - 54 \kappa_4^{(4)} \kappa_6^{(4)} + 48 \kappa_6^{(4)2} - 16 \kappa_6^{(4)3} + \\
& 12 \kappa_3^{(4)} (9 \kappa_4^{(4)2} - 10 \kappa_4^{(4)} \kappa_6^{(4)} + 4 \kappa_6^{(4)2})) p^4 + 32 \kappa_{15}^{(4)3} (135 \kappa_{16}^{(4)3} - 80 \kappa_3^{(4)3} - 27 \kappa_4^{(4)3} - \\
& 90 \kappa_{16}^{(4)2} (4 \kappa_3^{(4)2} + 3 \kappa_4^{(4)2} - 2 \kappa_6^{(4)2}) + 54 \kappa_3^{(4)2} \kappa_6^{(4)2} - 48 \kappa_4^{(4)2} \kappa_6^{(4)2} + 16 \kappa_6^{(4)3} (4 \kappa_3^{(4)2} - 15 \kappa_4^{(4)2} + 72 \kappa_6^{(4)}) - 12 \kappa_3^{(4)2} \kappa_6^{(4)2} \\
& (9 \kappa_4^{(4)2} - 10 \kappa_4^{(4)} \kappa_6^{(4)} + 4 \kappa_6^{(4)2}) + 6 \kappa_{16}^{(4)2} (52 \kappa_3^{(4)2} + 27 \kappa_4^{(4)2} + 8 \kappa_3^{(4)} (9 \kappa_4^{(4)} - 5 \kappa_6^{(4)}) - 36 \kappa_4^{(4)} \kappa_6^{(4)} + 16 \kappa_6^{(4)2})) p^4 + \\
& \kappa_{16}^{(4)2} (128 \kappa_3^{(4)4} p^4 + 27 \kappa_4^{(4)4} p^4 + 64 \kappa_3^{(4)3} (5 \kappa_4^{(4)} - 2 \kappa_6^{(4)}) p^4 - 72 \kappa_4^{(4)3} \kappa_6^{(4)} p^4 - 64 \kappa_4^{(4)2} \kappa_6^{(4)2} p^4 + 16 \kappa_6^{(4)2} p^4 + \\
& 12 \kappa_6^{(4)2} (9 \kappa_1^{(4)} + 8 \kappa_4^{(4)} p^4) - 8 \kappa_3^{(4)} (27 \kappa_6^{(4)2} \kappa_1^{(4)} - 18 \kappa_4^{(4)3} p^4 + 30 \kappa_4^{(4)2} \kappa_6^{(4)} p^4 - 24 \kappa_4^{(4)} \kappa_6^{(4)2} p^4 + 8 \kappa_6^{(4)3} p^4) + \\
& 12 \kappa_3^{(4)2} (2 \kappa_6^{(4)2} (12 \kappa_1^{(4)2} + 12 \kappa_4^{(4)2} + 12 \kappa_6^{(4)2}) + 12 \kappa_3^{(4)} (4 \kappa_4^{(4)2} + 4 \kappa_6^{(4)2}) + 12 \kappa_4^{(4)2} \kappa_6^{(4)2}) + 1728 \kappa_{15}^{(4)6} p^4 +
\end{aligned}$$

[illegible]

|  |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | 1 | $\frac{3 \kappa_1^{(2)}}{\begin{pmatrix} 4 \\ 4 \end{pmatrix} \begin{pmatrix} 4 \\ 4 \end{pmatrix} - \kappa_{15} + \kappa_{16}}$                                                                                                                                                                                                                                                                                                                                           | $\frac{1}{\begin{pmatrix} 4 \\ 4 \end{pmatrix} \begin{pmatrix} 4 \\ 4 \end{pmatrix} - \kappa_{15} + \kappa_{16}}$                                                                                                                                                                                                                                                                                                                                                                                                                |
|  | 3 | $\frac{4 \begin{pmatrix} 4 \\ 4 \end{pmatrix} \begin{pmatrix} 4 \\ 4 \end{pmatrix} \begin{pmatrix} 2 \\ 2 \end{pmatrix}}{\begin{pmatrix} 4 \\ 4 \end{pmatrix} \begin{pmatrix} 4 \\ 4 \end{pmatrix} \begin{pmatrix} 2 \\ 2 \end{pmatrix} - \kappa_{15} + \kappa_{16} + 2 \kappa_3 + 2 \kappa_3 \kappa_4 + \kappa_4 - \kappa_{16} (\kappa_3 + 2 \kappa_3 \kappa_4 - 2 (\kappa_3 + \kappa_4) \kappa_6 + 2 \kappa_{15} (2 \kappa_{16} - 3 \kappa_3 - 2 \kappa_4 + \kappa_6))}$ | $\frac{2 (12 \kappa_{15}^2 + 3 \kappa_{16}^2 + 6 \kappa_3^2 + 8 \kappa_3 \kappa_4 + 3 \kappa_4^2 - 4 \kappa_3 \kappa_6 - 4 \kappa_4 \kappa_6 + 2 \kappa_6^2 + 4 \kappa_{15} (2 \kappa_{16} - 4 \kappa_3 - 3 \kappa_4 + 2 \kappa_4 \kappa_6) - 8 \kappa_3 \kappa_6 - 4 \kappa_4 \kappa_6)}{(4 \kappa_{15} + \kappa_{16})^2 + 2 \kappa_3 + 2 \kappa_3 \kappa_4 + \kappa_4 - \kappa_{16} (\kappa_3 + 2 \kappa_3 \kappa_4 - 2 (\kappa_3 + \kappa_4) \kappa_6 + 2 \kappa_{15} (2 \kappa_{16} - 3 \kappa_3 - 2 \kappa_4 + \kappa_6))}$ |